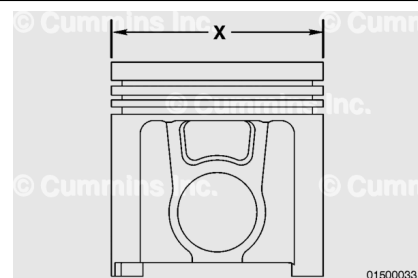


Trainings ▾

General Information

There are four pistons available for QSK45/60 engines. Two of these are ferrous cast ductile (FCD) pistons and are referred to as standard and cut back top land pistons. The other two types are steel pistons. The standard piston is used in QSK60 PowerGen engines built before 13 August 2012, all other QSK60 engines, and all QSK45 engines built before 08 April 2013. The cut back top land piston was introduced on all QSK60 PowerGen engines built after 13 August 2012 and all other engines built after 08 April 2013. The standard and cut back top land are also referred to as FDC pistons.

The two FCD pistons can be identified by the difference in the piston top land diameter when measured as shown in the illustration. The cut back top land piston has a smaller diameter than the standard piston.

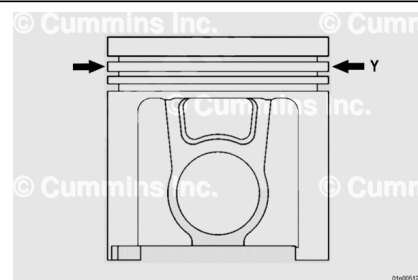


Approximate Top Land Diameter (X)

Piston Description	mm	in
Standard	156.96	6.18
Cut back top land	155.76	6.13

The other two types of piston are steel pistons, used on some field upfit engines. There are two variations of the steel piston, a standard piston and an offset piston. The offset piston is for production and rebuild upfit units.

The two steel pistons can be identified by the difference in the piston second land diameter when measured as shown in the illustration. The cut back second land piston has a smaller diameter than the standard piston.



Approximate Second Land Diameter (Y)

Piston Description	mm	in
Standard	157.92	6.22

Approximate Second Land Diameter (Y)		
Piston Description	mm	in
Offset	157.62	6.21

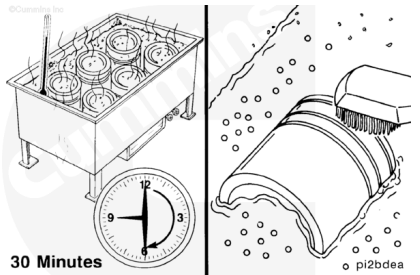
Preparatory Steps

- Remove the piston from the connecting rod. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html>)
- Remove and discard the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-047-tr.html>)

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Be sure the cleaning solvent is approved for ductile iron. Damage to the pistons can result if an unapproved solution is used.

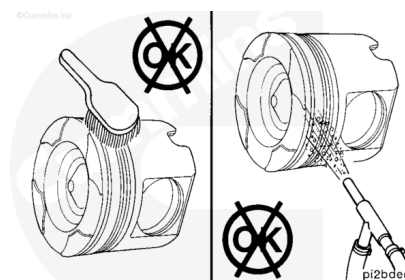
Allow the pistons to soak for a minimum of 30 minutes in a tank containing an approved cleaning solvent for ductile iron.

Use a hot, soapy solution and a non-metallic brush to remove carbon deposits.

⚠ CAUTION ⚠

Do not use a metal brush. A metal brush will damage the piston ring grooves. Do not use glass beads to clean the grooves. Walnut shell or plastic bead, Part Number 3822735, blasting can be used on ring grooves and on the dome or

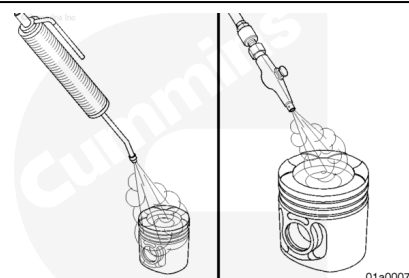
crown of the piston. Use the minimum effective pressure and do not concentrate the spray in one area for an extended period. The recommended blast pressure for plastic bead blasting is 276 kPa [40 psi].



Clean the pistons with walnut shells or plastic beads.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

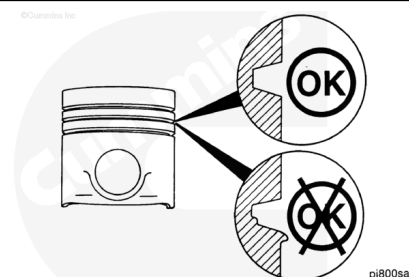


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

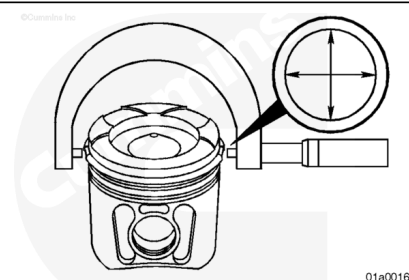
Use steam to clean the pistons and dry with compressed air.

Inspect the ring grooves. The pistons **must** be replaced if there is a visible ridge in the back of the groove, or if a lip has formed on the outer diameter of the groove.



Use piston ring groove wear gauge, Part Number 3824846, and a 152.4 to 177.8 mm [6 to 7 in] micrometer to inspect the top groove (compression ring).

Measure the top and second ring grooves (diameter over pins) when the piston temperature is at 21°C [70°F] in two places, 90-degrees apart, as shown.



Ring Groove Wear Limits for FCD Pistons (Diameter Over Pins)

	mm		in
Top Ring	159.10	MIN	6.264
Second Ring	159.41	MIN	6.276

Ring Groove Wear Limits for Steel Piston (Diameter Over Pins using 3.4 mm pin)

	mm		in
Top Ring	156.436	MIN	6.159

For the second ring groove, do **not** use the diameter over pin method.

Measure the second ring groove height.

Second Ring Groove Height for Steel Piston

mm		in
3.09	MAX	0.122
3.07	MIN	0.121

If the piston is **not** within specifications, the piston **must** be replaced.

Measure the oil ring groove.

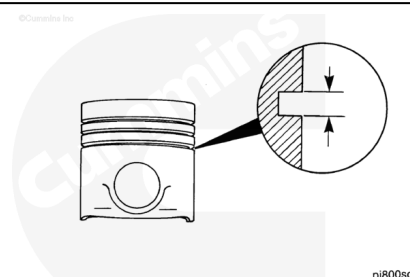
Piston Oil Ring Groove Width for FDC Piston

mm		in
4.788	MIN	0.1885
4.851	MAX	0.1910

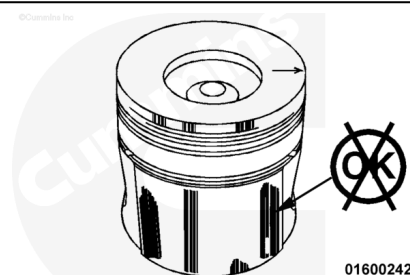
Piston Oil Ring Groove Width for Steel Piston

mm		in
4.03	MIN	0.1586
4.05	MAX	0.1594

If the piston is **not** within specifications, the piston **must** be replaced.



Inspect each piston for scratches or indications of scuffing. Refer to the Parts Reuse Guidelines, Bulletin 3810303.

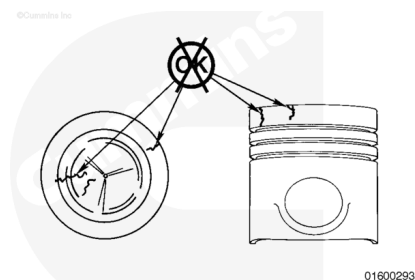


Use crack detection kit, Part Number 3375432, or equivalent, to check for cracks on top and in the pin bore of the piston.

The piston **must** be replaced if there is a crack in the piston pin bore.

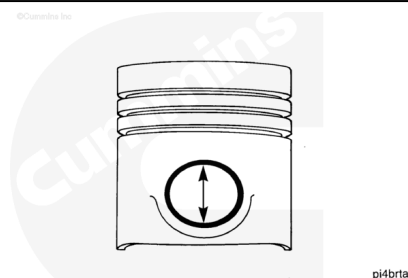
The piston **must** be replaced if there are any cracks on the top or the rim of the piston.

Note : On the steel piston, check specifically for cracks around the welded areas of the piston.



The piston measurement is to be recorded at room temperature of approximately 21° C [70° F].

Measure and record the inside diameter of the piston pin bore at several points as illustrated.



Piston Pin Bore Inside Diameter for FCD Piston

mm		in
65.00	MIN	2.559
65.10	MAX	2.563

Piston Pin Bore Inside Diameter for Steel Piston

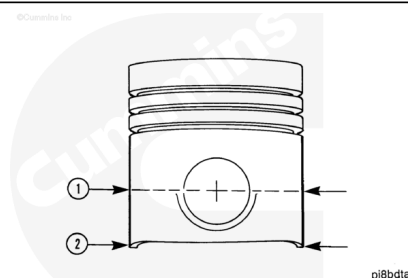
mm		in
65.09	MIN	2.5626
65.11	MAX	2.5633

Note : The pin bore of the steel piston is profiled.

If the piston pin bore is **not** within specifications, the piston **must** be replaced.

Measure the piston skirt at the centerline of the piston pin bore (1).

Measure the piston skirt at the bottom of the skirt (2). The measurement **must** be taken approximately 15 mm [0.625 in] from the bottom of the skirt.



Piston Skirt Outside Diameter for FCD Piston

	mm		in
(1)	158.631	MIN	6.245
	158.671	MAX	6.247
(2)	158.480	MIN	6.239
	158.520	MAX	6.241

Piston Skirt Outside Diameter for Steel Piston

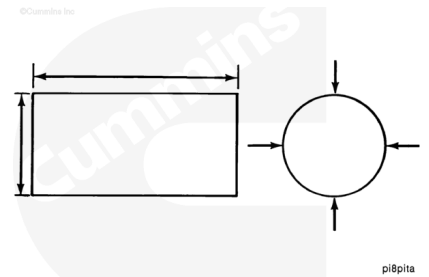
	mm		in
(1)	158.641	MIN	6.246
	158.671	MAX	6.247
(2)			

If the piston is **not** within specifications, the piston **must** be replaced.

Measure the piston pin outside diameter and length.

Piston Pin

	mm		in
Outside Diameter	64.997	MIN	2.5589
	65.003	MAX	2.5591
Length	118.56	MIN	4.668
	118.80	MAX	4.677



If the piston pin is **not** within specifications, the piston pin **must** be replaced.

Finishing Steps

- Install the piston rings. Refer to Procedure 001-047 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-047-tr.html)
- Install the piston to the connecting rod. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)